

Planetary Protection Biosafety Level 4 Geological Sample Isolation

1. Pristine Geological Containment

Sub-surface Martian soil and rock cores are sealed within double-walled brazed titanium canisters under negative pressure vacuum seals at the drill head.

2. Dry Heat Microbial Sterilization

Sample canister exteriors undergo dry heat microbial reduction (DHMR) at 135 degrees Celsius for 42 hours before transit through cleanroom airlocks to eliminate cross-contamination.

3. Glovebox Raman Spectrometry

Non-destructive Raman, XRF, and infrared spectroscopy inspect sample mineralogy within a glovebox filled with ultra-pure argon atmosphere at positive pressure.